

VYVASWATH KALVA

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EDUCATION

Carnegie Mellon University

Master of Science, Mechanical Engineering - Research | Current GPA: 4.0/4.0

May 2027

Pittsburgh, PA

Indian Institute of Technology

Bachelor of Technology, Mechanical Engineering | GPA: 9.0/10

July 2024

Indore, India

Relevant Coursework: Spacecraft Attitude Dynamics and Control, Instrumentation & Control, Numerical Methods

GRADUATE COURSEWORK

Robot Dynamics and Analysis, Modern Control Theory, Machine Learning and AI, Intro to Robot Learning, Optimal Control and Reinforcement Learning

EXPERIENCE

Carnegie Mellon University

Graduate Researcher (Advisor: Prof. Hartmut Geyer)

Oct 2025 – Present

Pittsburgh, PA

- Developing sim-to-real deep RL locomotion controllers for a powered transfemoral prosthesis.
- Building a Python-based real-time control stack from scratch for hardware deployment.

Carnegie Mellon University

Course Assistant - Mechanical Systems Experimentation (24-452)

Sept 2025 – Dec 2025

Pittsburgh, PA

- Conducted laboratory sessions and office hours for dynamic systems and controls experiments, including system identification and PID control.
- Graded lab reports and provided feedback on experimental analysis.

Indian Institute of Science

Research Intern

Jul 2024 – Mar 2025

Bengaluru, India

- Redesigned and optimized a modular articulated robot, enabling reliable locomotion in inclined pipes.
- Modeled and simulated robot dynamics on MATLAB, producing simulations that guided development of control strategies.

Zebu Intelligent Systems

Mechanical Intern

May 2023 – Jul 2023

Hyderabad, India

- Designed and analyzed structural components for a surveillance helicopter, contributing to a working prototype.
- Designed the swashplate mechanism, analyzed transmission issues, and integrated Pixhawk Cube with ArduPilot for flight readiness.

PROJECTS

LocoMimic: Learning Locomotion Control | Intro to Robot Learning (16-831) Project

Jan 2026 - Apr 2026

- Evaluated deep reinforcement learning algorithms for humanoid locomotion, utilizing a GPU-accelerated framework to train a 29-DOF Unitree G1 robot to track retargeted motion.
- Resolved critical training instabilities in Vanilla Soft Actor-Critic models during high-dimensional control tasks by implementing MeanSAC.
- Trained MeanSAC policy, successfully eliminating mid-training reward collapses and achieving higher mean episode returns than the baseline implementation.

TECHNICAL SKILLS

Programming: Python, MATLAB

Frameworks: Gymnasium, PyTorch, ROS 2

Technical: Mujoco, SolidWorks, Fusion 360, Ansys Mechanical, ADAMS, Onshape

Other: Linux, Git, Arduino IDE, $\LaTeX$

PATENTS

Robotic Joint Apparatus with Multiple Degrees of Freedom and Method Thereof

Patent No. 587636, Grant Date: 24th April 2026

A Gesture-controlled SMA-actuated Robotic Gripper System and Method Thereof

Patent No. 568766, Grant Date: 17th July 2025